

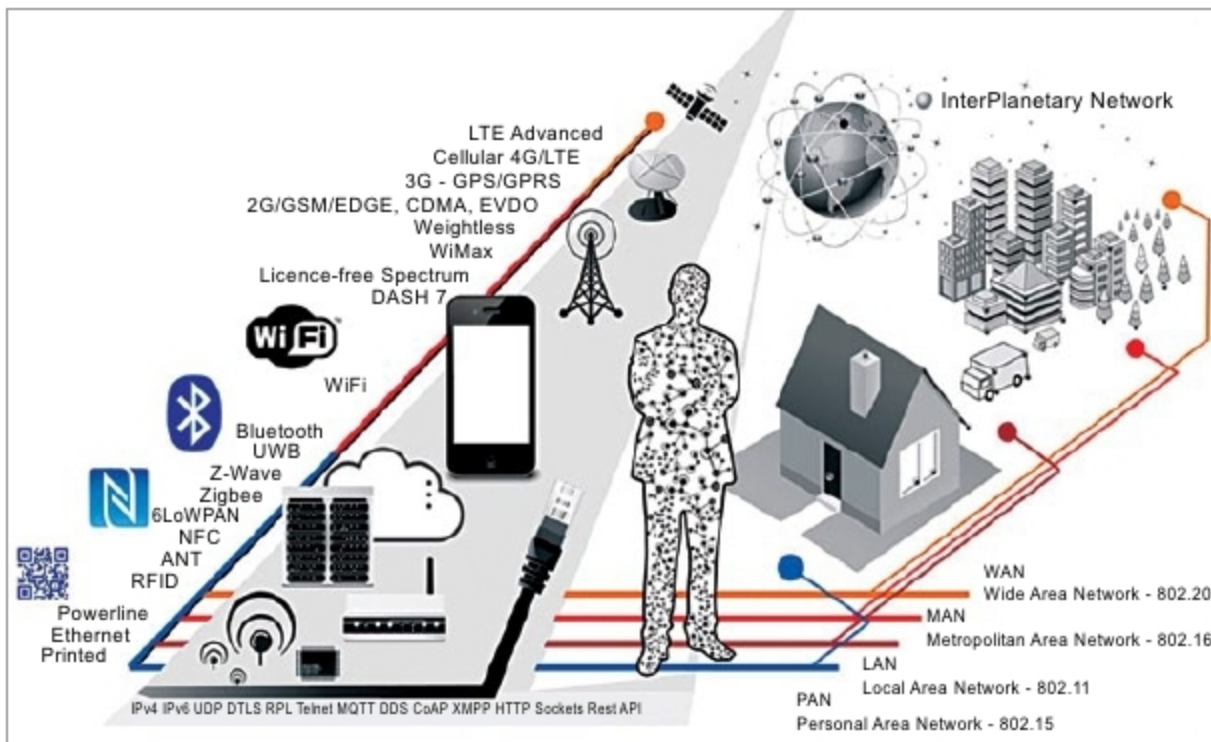


Fig. 2: Wireless connectivity options for the IoT (Credit: Postscapes)

will become crucial to meet all these requirements.

End applications drive growth in different sectors, including healthcare, IT, telecommunications and automotive. Both short- and long-range technologies are addressed as per the need of the business. Bluetooth and Wi-Fi technologies are for short range in unlicensed bands, and longer-range LTE, NB-IoT, and LTE Cat M1 and M0 in licensed bands.

M2M communication in the IoT requires reliable data collection and uninterrupted data transmission. There are primarily three types of wireless communication technologies helping IoT devices in communicating data. These are:

- LTE-M and NB-IoT networks that were developed to support IoT devices even in the most challenging environments, such as underground and deep inside buildings. These have extended battery life and enhanced coverage. Licensed spectrum eliminates the potential for signal interference and dedicated spectrum optimised for the IoT with carrier-grade security.

- Global SIMs that deploy and control connected devices easily and quickly, and automatically and remotely connect to the network for provisioning IoT assets worldwide. These are based on GSM standard for high-speed wireless delivery. There is a standard interface for deploying and connecting mobile and fixed assets. Simple and cost-effective device man-

agement can be done with integrated SIMs, which have global roaming in over 200 countries and territories.

- Satellite connectivity that gives greater control of assets, no matter how distant and remote. It manages high-bandwidth demands in mobile and fixed locations. Three solutions built to fit needs include dual-mode, satellite-only and satellite-backhaul. Their usage-based pricing models have greater cost efficiencies.

Staying connected is important for businesses for processing payments to analysing data, as business continuity can be compromised in network failure in the event of Internet outages. Choosing and integrating the right components for a complete IoT solution is a challenge.

The Industrial Internet Consortium (IIC) and OpenFoG Consortium are trying to accelerate the adoption of the IIoT, fog and edge computing, to address bandwidth, latency and communications challenges associated with the IoT, 5G and artificial intelligence (AI) applications. Their mission is to create and validate a framework for efficient and secure information processing between endpoints, cloud and services.

## Growth opportunities for the IoT - T&M industry

With the IoT coming into prevalence, the world is becoming increasingly connected, and companies providing IoT testing services are emerging. From security systems and appliances

to new and emerging devices, all are getting connected over the Internet. The growing market for the IoT has created a number of business opportunities.

Very strict coverage requirements call for smart testing in IoT applications where devices have to operate under difficult coverage conditions. It requires comprehensive testing of the final device design under real-world conditions in various operational modes. Opportunities in IoT products testing will grow, and companies that focus on quality will ultimately be the brands that win.

With over-the-air (OTA) firmware updates through connected devices, it is possible to perform baseline functional and compliance testing, and ship products as soon as possible. This also helps fix any issues that pop up through regular updates. Reliable testing of interference, conformance with standards and regulations, latency and power consumption play an important role here. This reduces time-to-market, increases demand and improves quality of products.

The technology shift towards 5G will require a massive amount of testing for connected cars and autonomous vehicles, too.

## IoT solutions for connected businesses

The IoT has the power to transform the businesses of any industry. It can improve employee efficiency, deliver better customer experiences, and track supply chain, assets and operations. IoT networks connect people, processes and applications so that businesses can quickly adapt to changing conditions. Given below are a few IoT solutions required by every industry.

- Cloud hosting provides simple scalability for IoT apps.

- IoT solution management answers needs with end-to-end support for selling, storing and managing IoT solutions. This includes connectivity services, devices, applications, connectivity management, analytics platforms, billing, and device movement control for asset and object tracking.

- IoT security management secures IoT environments. It helps identify things in the connected world and ensure that these are used only by authorised users, devices and applica-